Functions - Part 4: Exponential and Logarithmic Functions

Course: Pre-Calculus-I

AH Sheikh

26CS(Evening) - Fall 2026

Outline

1 Logarithmic Functions

2 Comparison and Summary

What is a Logarithm?

Definition

The **logarithm** base a of a positive number x is the exponent to which a must be raised to obtain x:

$$\log_a x = y \iff a^y = x$$

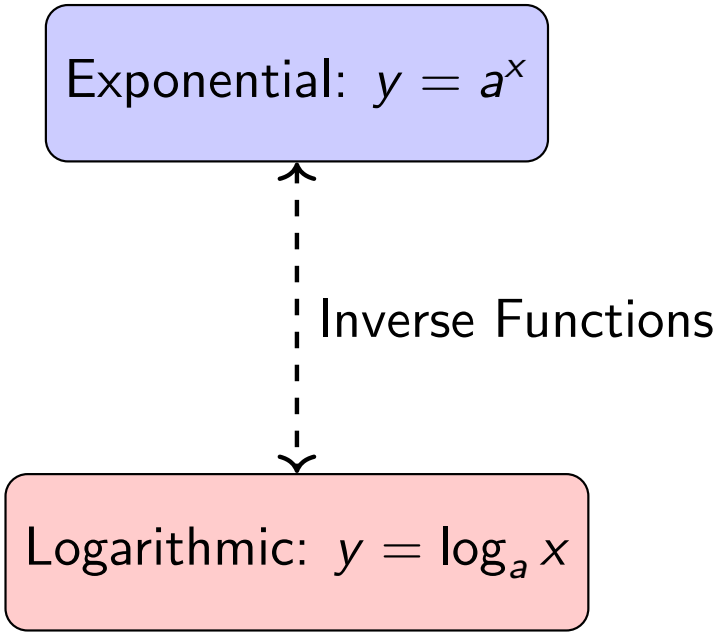
where $a > 0$, $a \neq 1$, and $x > 0$.

Key Idea: Logarithms are the **inverse** of exponentials.

Common Logarithms

- **Common logarithm:** $\log x = \log_{10} x$
- **Natural logarithm:** $\ln x = \log_e x$

Relationship Between Exponential and Logarithmic Functions



Key Relationship: $y = a^x \iff x = \log_a y$

- The graph of $y = \log_a x$ is the **reflection** of $y = a^x$ across the line $y = x$.
- Domain of $\log_a x$: $(0, \infty)$.
- Range of $\log_a x$: $\mathbb{R}$.

Properties of Logarithmic Functions

Laws of Logarithms

For $a > 0$, $a \neq 1$, and positive real numbers x, y:

$$\log_a(xy) = \log_a x + \log_a y$$

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

$$\log_a(x^n) = n \log_a x$$

$$\log_a(1) = 0$$

$$\log_a(a) = 1$$

$$\log_a(a^x) = x$$

$$a^{\log_a x} = x$$

How to Graph Logarithmic Functions

Step-by-Step Method:

1. **Identify the base** a and the parent function $y = \log_a x$.
2. **Determine the shape:**
   - $a > 1$: Increasing function.
   - $0 < a < 1$: Decreasing function.
3. **Find key points:**
   - x-intercept: $(1, 0)$ since $\log_a 1 = 0$.
   - Additional points: $(a, 1)$, $(1/a, -1)$, $(a^2, 2)$.
4. **Draw the vertical asymptote:** $x = 0$ (the y-axis).
5. **Plot points and connect** with a smooth curve.
6. **Apply transformations** if the function is in the form $y = \log_a(x - h) + k$.

Example 1: Graph of $f(x) = \log_2 x$

Key Features:

- Base $a = 2 > 1$: Increasing.
- x-intercept: $(1, 0)$.
- Vertical asymptote: $x = 0$.
- Domain: $(0, \infty)$.
- Range: $\mathbb{R}$.
- Passes through $(2, 1)$ and $(4, 2)$.

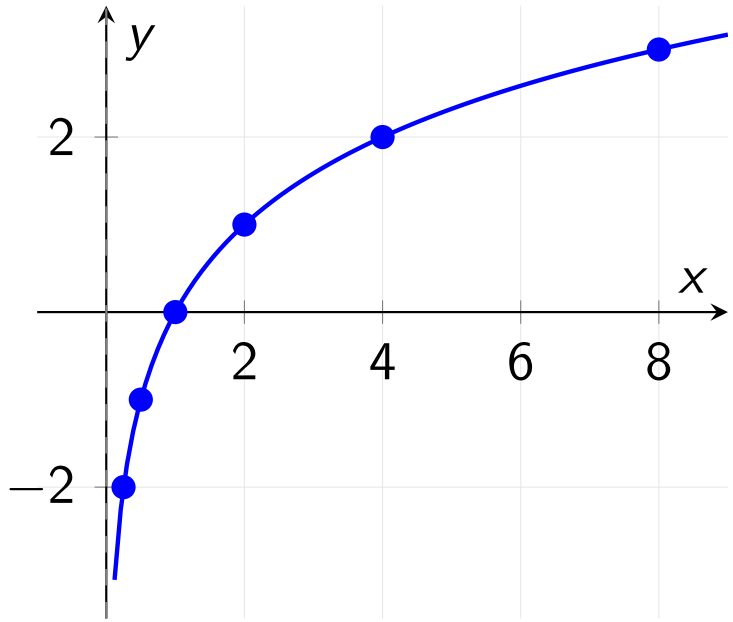


Table of Values:

x	$y = \log_2 x$
0.25	-2
0.5	-1
1	0
2	1
4	2

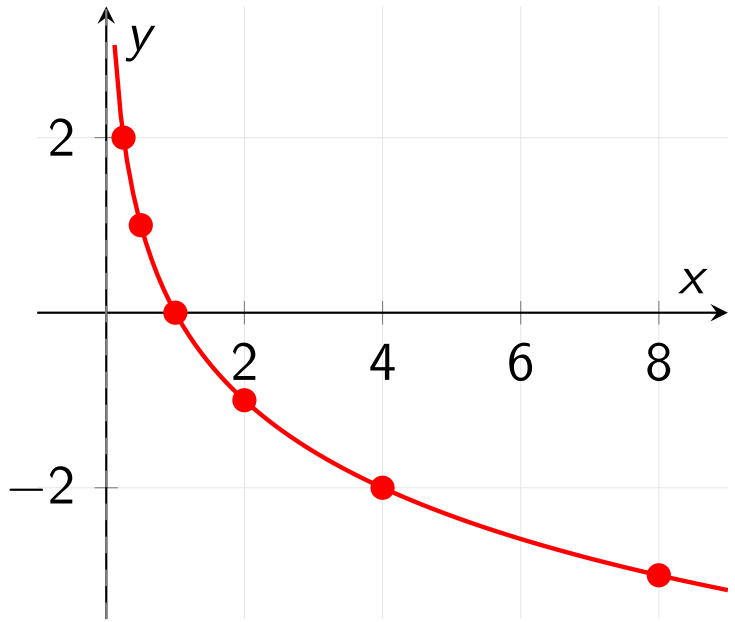
Example 2: Graph of $f(x) = \log_{1/2} x$

Key Features:

- Base $a = \frac{1}{2} < 1$: Decreasing.
- x-intercept: $(1, 0)$.
- Vertical asymptote: $x = 0$.
- Domain: $(0, \infty)$.
- Range: $\mathbb{R}$.

Table of Values:

x	$y = \log_{1/2} x$
0.25	2
0.5	1
1	0
2	-1
4	-2
8	-3



Example 3: Graph of $f(x) = \ln x$

Key Features:

- Natural logarithm: base $e \approx 2.718$.
- x-intercept: $(1, 0)$.
- Vertical asymptote: $x = 0$.
- Domain: $(0, \infty)$.
- Range: $\mathbb{R}$.
- Inverse of $y = e^x$.

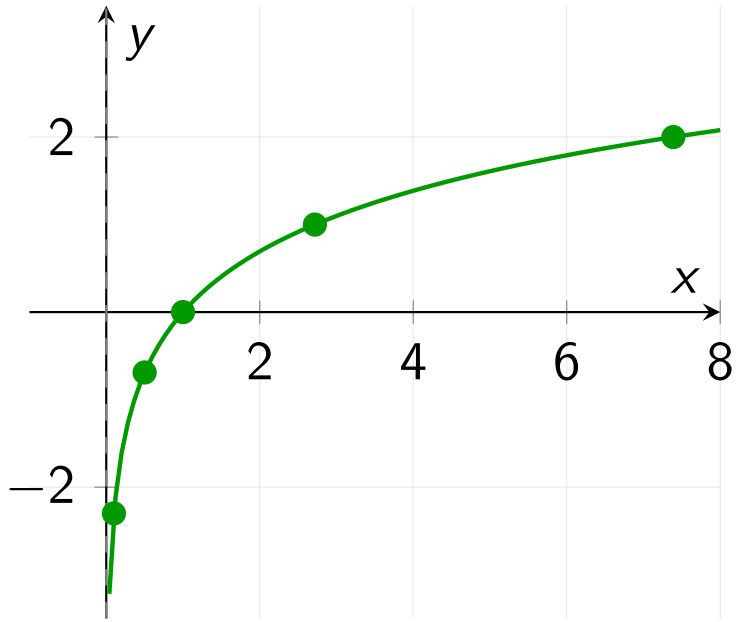


Table of Values:

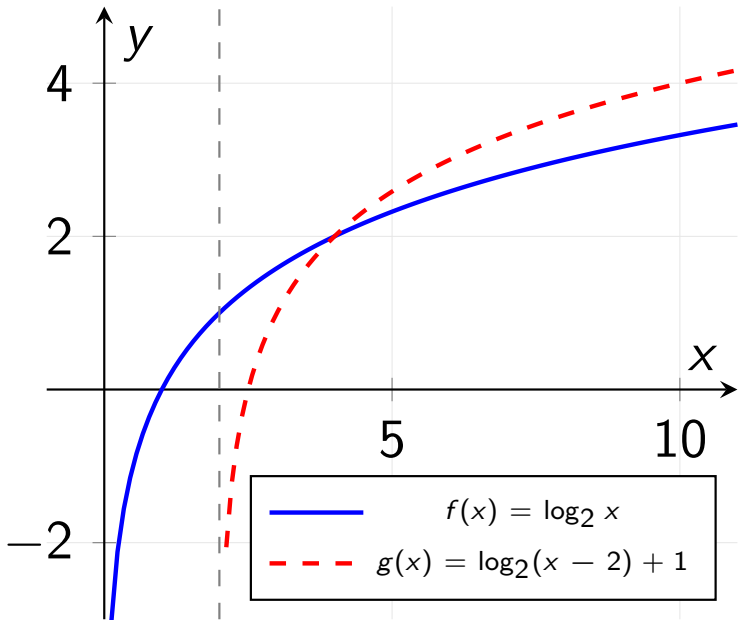
x	$y = \ln x$
0.1	-2.30
0.5	-0.69
1	0
e	1

Example 4: Transformations of Logarithmic Functions

Graph $f(x) = \log_2 x$ **and** $g(x) = \log_2(x-2) + 1$.

Transformations:

- Start with $f(x) = \log_2 x$.
- $(x-2)$: Shift **right** by 2 units.
- $+1$: Shift **up** by 1 unit.
- New asymptote: $x = 2$.
- New x-intercept: $(2.5, 0)$.



Key Points:

x	$\log_2 x$	$\log_2(x-2)+1$
2.5	-	0
3	1.58	1
4	2	2
6	2.58	3
10	3.32	4

Example 5: Graph of $f(x) = -\ln(x+1) + 2$

Transformations:

- Start with $y = \ln x$.
- $(x+1)$: Shift **left** by 1 unit.
- $-$ sign: Reflect over x-axis.
- $+2$: Shift **up** by 2 units.
- Asymptote: $x = -1$.
- Domain: $(-1, \infty)$.

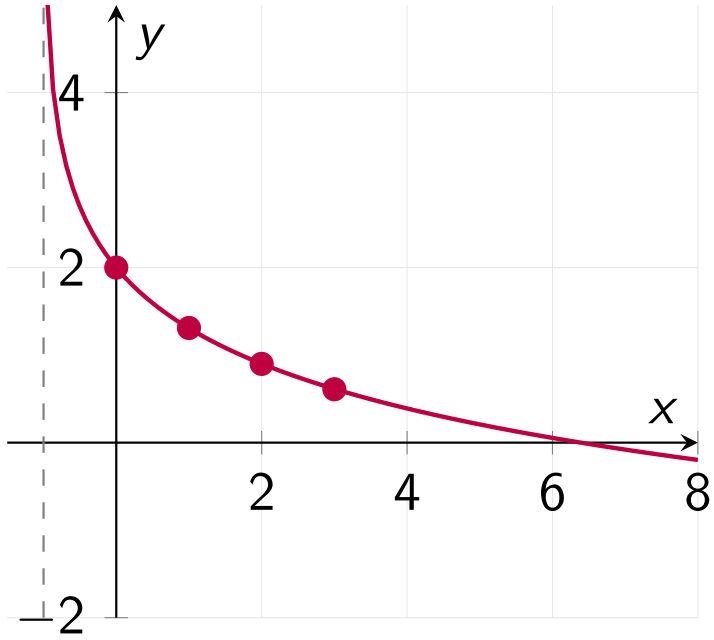


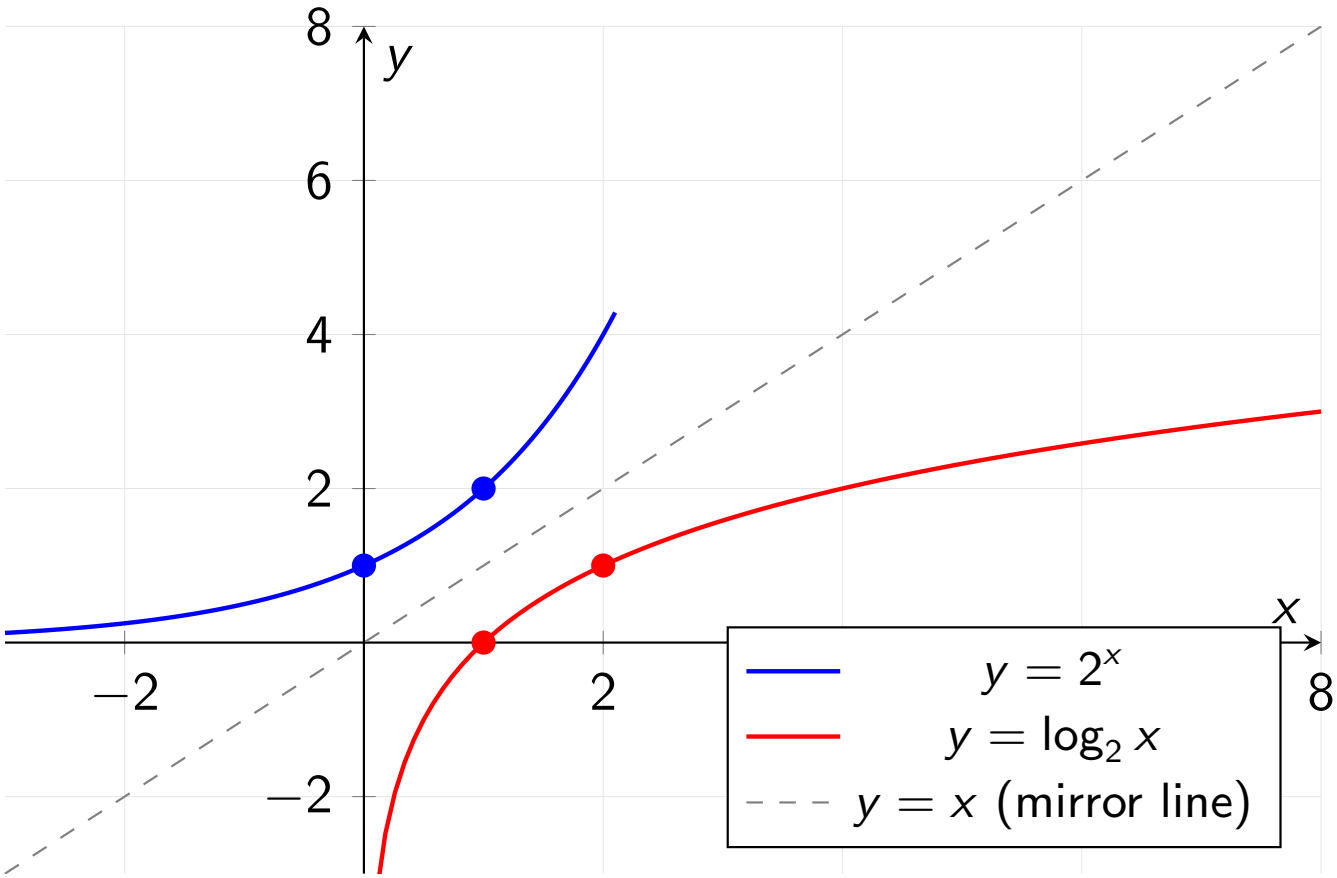
Table of Values:

x	$-\ln(x+1)+2$
0	2
1	1.31
2	0.90
3	0.61
e-1	1

Exponential vs. Logarithmic Functions

Feature	**Exponential** $y = a^x$	**Logarithmic** $y =$
Definition	$a^y = x$ inverse relation	$\log_a x = y$
Domain	$(-\infty, \infty)$	$(0, \infty)$
Range	$(0, \infty)$	$(-\infty, \infty)$
Asymptote	Horizontal: $y = 0$	Vertical: $x = 0$
Key Point	$(0, 1)$	$(1, 0)$
Growth $(a > 1)$	Increasing	Increasing
Decay $(0 < a < 1)$	Decreasing	Decreasing
Inverse of	Logarithmic	Exponential

Inverse Relationship: Graphical View



The graphs are reflections of each other across the line $y = x$.

Common Mistakes to Avoid

Mistake 1: Domain of Logarithms

$\log_a x$ is only defined for $x > 0$. Always check the argument!

- $\log_2(x-3)$ requires $x - 3 > 0 \implies x > 3$.

Mistake 2: Confusing the Laws

$$\log_a(x+y) \neq \log_a x + \log_a y$$

The sum law is for **products**, not sums!

Mistake 3: Asymptotes

- Exponential: horizontal asymptote (usually $y = k$).
- Logarithmic: vertical asymptote (usually $x = h$).

Applications

Exponential Functions:

- Population growth: $P(t) = P_0 e^{kt}$
- Compound interest: $A = P\left(1 + \frac{r}{n}\right)^{nt}$
- Radioactive decay: $N(t) = N_0 e^{-kt}$
- Cooling (Newton's Law): $T(t) = T_s + (T_0 - T_s)e^{-kt}$

Logarithmic Functions:

- pH scale: $\text{pH} = -\log[H^+]$
- Richter scale: $M = \log\left(\frac{I}{I_0}\right)$
- Decibels: $L = 10\log\left(\frac{I}{I_0}\right)$
- Solving exponential equations: log both sides

Summary

Key Takeaways

- **Exponential functions** $f(x) = a^x$:
  - Domain: $\mathbb{R}$, Range: $(0, \infty)$.
  - Horizontal asymptote at $y = 0$.
  - Growth if $a > 1$, decay if $0 < a < 1$.
- **Logarithmic functions** $f(x) = \log_a x$:
  - Domain: $(0, \infty)$, Range: $\mathbb{R}$.
  - Vertical asymptote at $x = 0$.
  - Inverse of exponential functions.
- **Graphing:** Use key points, asymptotes, and transformations.
- **Transformations** work the same way as before:
  - $f(x - h) + k$ shifts right/left and up/down.